

Affest: Verifiable Cross-Chain Portfolio Automation

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Affest
Built for BUIDL CTC 2026 Fall

Abstract. Affest is a non-custodial portfolio automation layer built on Creditcoin and powered by Attestcoin. It converts verified events from external chains into guarded portfolio actions. Users define allocation rules, fund a self-owned vault, and retain final approval over every transaction. Affest gives people and AI agents a deterministic way to react to cross-chain data without sharing private keys or trusting an off-chain executor.

1. Introduction

On-chain portfolios operate across fragmented networks, yet most automation tools can only observe and act within one chain. Reacting to an event elsewhere usually requires a centralized API, a trusted relay, or a bot with broad wallet permissions. These systems are difficult to audit and may expose users to custody, execution, and data-integrity risks. Affest separates observation, proof, policy evaluation, and execution so that each stage can be inspected before capital moves.

2. The Affest Protocol

An Affest strategy records a target allocation, a source-chain trigger, an optional cooldown, and an expiry time. For example, a user may instruct Affest to maintain 70 percent WTCTC and 30 percent risk exposure after a qualifying WETH signal is verified on Ethereum Sepolia. Until that signal is proven, the strategy remains waiting. Once verified, Affest compares the vault's current holdings with its target and prepares the minimum rebalance required to restore the policy.



Figure 1. Affest's approval-gated cross-chain flow.

3. System Components

Strategy vaults hold supported assets under user-owned rules. The **strategy engine** validates allocations, contracts, triggers, cooldowns, and expiry conditions. The **Attestcoin proof layer** establishes that a referenced source transaction and event occurred. The **Affest MCP** lets users and AI agents inspect portfolios, draft and validate strategies, diagnose vaults, preview rebalances, and prepare unsigned transactions through natural language. The **approval guard** prevents final execution without the vault owner's signature.

4. Security and Control

Affest does not receive private keys and the MCP cannot sign or independently move funds. A proposal must match the saved on-chain strategy, current vault state, and a verified trigger. Idempotency controls reduce duplicate requests, while invalid tokens, mismatched vault ownership, insufficient funding, missing proofs, and expired policies return structured diagnostics. This design makes automation observable and approval-gated rather than opaque or fully autonomous.

5. Prototype and Direction

The current prototype runs on Creditcoin CC3 Testnet and uses Ethereum Sepolia as a source chain. It supports live portfolio reads, deterministic strategy creation, vault diagnostics, trigger tracking, proof-readiness checks, rebalance previews, and unsigned approval preparation. At the time of writing, funded strategies 6 and 7 validate the 70/30 policy flow and are ready at the vault level while awaiting their configured cross-chain trigger. Future work includes additional source chains, production-grade asset integrations, richer policy conditions, and reusable strategy templates for both human and agent workflows.

Affest automates the strategy, verifies the signal, and leaves execution under the user's control.